Iheb Belhadj

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Summary

Software engineer specializing in backend systems, data pipelines, and agentic AI. Experienced in building cloud-native applications, multi-agent systems, and scalable infrastructure for data-driven solutions.

Technologies

Languages: TypeScript, Python, JavaScript, Java, C#, C, SQL, PHP, Bash
Web/Fullstack: Next.js, React, Astro.js, Angular, Vue.js, Express.js, Spring, Flask, FastAPI
Databases: PostgreSQL, Oracle DB, MySQL, MongoDB, Redis, MinIO, AWS RDS, S3, Vercel Blob, Neon Postgres
Data Eng & AI: Airflow, Airbyte, dbt, Kafka, Spark, LangChain, LangGraph, MLflow, Snowflake, Databricks
Infrastructure: Docker, Kubernetes, Ansible, Vagrant, Jenkins, GitHub Actions, AWS
Testing Vitest, Playwright, Jest, Jasmine, Pytest, SonarQube

Experience

Freelance Job, K-Rose – Tunis,TN feb 2026 – may 2026

- Lead the architecture and development of a B2B e-commerce platform for the floral industry **kairouanrose.com** , integrating a headless CMS as the backbone for both business operations (order, delivery, and product management) and content delivery (dynamic SEO rule sets, automated sitemap, and CMS-managed pages compiled as SSGs) — achieving Lighthouse scores of 98 SEO and 100 Accessibility.
- Implemented role-based access control with row-level ownership enforcement, mapping Clerk authentication to CMS domain entities across two independent layers
- Structured error logging via next-axiom with contextual metadata to inspect production errors
- Configured tag-scoped ISR cache invalidation tied to CMS publish events, with bulk-import-safe revalidation suppression
- Built a field-level audit logging system capturing structured diffs, actor metadata, and request context across all CMS collections
  - Technologies used: Next.js , Payload CMS, React , Typescript, Tailwind, Vercel, Neon Postgres, Next-axiom, Clerk, Vitest, Playwright, Vercel Blob

Data engineering Consultant, Talan – Tunis,TN Oct 2025 – currently

- Lead the migration of a CRM data integration project to dbt and snowflake
- Developed a native tool in snowflake for writing xml ingestion code based on a given xsd schema and a defined config by the end user, eliminating the need for external ingestion tools
- Working closely with GRDF to fix detected production anomalies in their legacy data pipelines.
- Contributing to the development of an agentic AI system to automate the migration of Informatica mappings to dbt models, drastically reducing conversion time.
  - Technologies used: Python, FastAPI, React, LangGraph, Redis, dbt, Docker, Informatica, VTOM, Oracle DB, Snowflake

Software Engineering Intern, Talan – Tunis,TN Feb 2025 – Oct 2025

- Integrated a multi-agent system with LangGraph for dynamic SQL and chart generation, enabling no-code data exploration.
- Built a real-time CDC pipeline refreshing charts live, enabling instant monitoring through the AI first client interface.
- Developed an end-to-end data pipeline with MinIO, Nessie, Airflow, and dbt to manage ingestion, transformation, and ML forecasting.
- Secured the platform with Keycloak IAM using role- and group-based access control.
- Implemented a caching system through Redis reducing network I/O calls for the agentic system for context retrieval.
  - Technologies used: Python, FastAPI, Next.js, LangGraph, LangChain, Redis, MLflow, MinIO, Nessie, Trino, Airflow, dbt, PostgreSQL, Keycloak, Docker, Nginx

Freelance Job, Nexits – Paris, FR (Remote) Dec 2024 - Jan 2025

- Architected a CMS solution with Payload CMS and a custom block-based page builder, enabling no-code content updates for the landing site.
- Built an instantaneous preview system using Astro.js for dynamic component rendering with minimal JS.
- Designed a dual rendering strategy (SSR for preview, SSG for production), achieving 100 performance, 95 accessibility, and 100 SEO score on Lighthouse.
  - Technologies used: Astro.js, Payload CMS, React, TypeScript, Vercel, SQLite, Turborepo

Full-stack Developer, Nexits – Paris, FR (Remote) Aug 2023 - Sep 2024

- Implemented product state historization with audit trail, enabling edits, rollbacks, and time-travel of product data.
- Contributed to migration from legacy stack to Next.js, tRPC, and Turborepo monorepo with AWS deployment.
  - Technologies used: Next.js, React, TypeScript, Node.js, tRPC, PostgreSQL, AWS (EC2, ECR, S3, RDS), Docker, GitHub Actions, Turborepo

Freelance instructor (Weekends), ENET'Com Junior Entreprise – Sfax, TN Oct 2023 - Nov 2023

- Conducted workshops over the weekends on web development using the MERN stack (MongoDB, Express.js, React, Node.js) teaching more than 30 junior club members.

Software Engineering Intern, Nexits – Paris, FR (Remote) June 2023 - Aug 2023

Projects

- **Playpex (Link):** Torrent streamer in Node.js with backend wrapper using MSE API for seamless browser playback. Angular frontend, Electron desktop app to bundle the engine and the frontend, and Ionic mobile controller.
- **OpsForge (Link):** A self-contained local DevOps environment simulating production using Vagrant and Ansible. Includes a private Docker registry, Jenkins CI/CD, monitoring stack with Prometheus/Grafana, and SonarQube.
  - Technologies used: Ansible, Vagrant, Docker, Jenkins, Prometheus, Grafana, SonarQube, Bash
- **EldSync (Link):** Contributed to the frontend development of a Java microservices platform, building Angular components with NgRx for state management, integrated with Spring Boot backend services.
- **Portfolio (Link):** Built on Next.js with MDX-based content management, lazy loading, and server components. Integrated a LangChain agent that answers questions about my experience. Achieved 100 performance and best practices scores on Lighthouse.

Education

Software Engineering Master, ESPRIT Sept 2022 – Sep 2025

Pre-engineering Bachelor, IPEIM Sept 2020 – Juin 2022